

5. Which row in the table gives the gravitational potential energy, the elastic potential energy and the kinetic energy of a bungee jumper during the first fall? Air resistance is negligible.

		Gravitational potential energy/ kJ	Elastic potential energy/ kJ	Kinetic energy/kJ
A	top	120	0	0
	middle	60	10	50
	bottom	0	120	0
B	top	120	0	0
	middle	60	30	30
	bottom	0	60	60
C	top	120	0	0
	middle	60	30	60
	bottom	0	120	0
D	top	120	0	0
	middle	60	60	0
	bottom	0	120	0

Ans: A

At the top: There is only GPE. Hence, EPE and KE equal to 0.

At the bottom: All GPE is lost and converted to EPE as it is momentarily at rest, hence KE = 0. Option B is wrong.

At the middle: GPE lost is converted to gain in EPE and KE. The bungee jumper is moving, so KE cannot be 0. Option D is wrong

The total energy stays constant at 120 kJ. Option C is wrong.

Only Option A gives the correct combinations of energy at the different positions.

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6 An escalator is 60 m long and lifts passengers through a vertical height of 30 m as